

Underfitting & Overfitting

In machine learning a model is provided time and data to produce his prediction for the upcoming results ,however, one of the many challenges that might occur are underfitting and overfitting.

Underfitting

Underfitting is a term that is used to describe a state where the model is not able to pick up on the general pattern of the data and produce inaccurate predictions as a result.

Usually due to :

- The lack of time , iteration or data provided for the model to compute.
- Choose a model that is not suitable, like a linear regression model for a polynomial function of data.

Overfitting

In contrast with underfitting , overfitting describes a model that is too accurate, following every pattern in the provided data including noises in samples, which will result in accurate predictions.

Usually due to :

- Models with high complexity with excessive number of parameters that misread the pattern of the data.
- Insufficient amount of data that results in making the model read the random samples as real and related data following the pattern.